

Tutorial 5 – Week 5

System Response

MTE 252

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Question 2.4-17

The unit impulse response of an LTIC system is

$$h(t) = [2e^{-3t} - e^{-2t}]u(t)$$

Find this system's (zero-state) response $y(t)$ if the input $x(t)$ is:

- (a) $u(t)$
- (b) $e^{-t}u(t)$

Use the convolution table (Table 2.1) to find your answers.

Solution 2.4-17 a)

impulse Response $h(t) = [2e^{-3t} - e^{-2t}] u(t)$

input $x(t) = u(t)$

Zero state response $y(t) = x(t) * h(t)$

↓

$$y_{zsr}(t) = u(t) * [2e^{-3t} - e^{-2t}] u(t)$$

↓

$$y_{zsr}(t) = \underbrace{2[e^{-3t} u(t) * u(t)]}_{(1)} - \underbrace{[e^{-2t} u(t) * u(t)]}_{(2)}$$

(1)

(2)

final answer: $\frac{2}{3}(1 - e^{-3t})u(t) - \frac{1 - e^{-2t}}{2}u(t) = \left[\frac{1}{6} + \frac{1}{2}e^{-2t} - \frac{2}{3}e^{-3t}\right]u(t)$

Now we look at the convolution table to find the convolution formula corresponding to our signals.

Row 2 seems to match.

Row 2 says if our signals are in the forms of: $e^{kt} u(t) * u(t)$

We should use this formula: $\frac{1 - e^{kt}}{-k} u(t)$

① $2[e^{-3t} u(t) * u(t)]$, $k = -3 \rightarrow 2\left(\frac{1 - e^{-3t}}{-(-3)}\right)u(t) = \frac{2}{3}(1 - e^{-3t})u(t)$

② $[e^{-2t} u(t) * u(t)]$, $k = -2 \rightarrow \frac{1 - e^{-2t}}{-(-2)}u(t) = \frac{1 - e^{-2t}}{2}u(t)$



Solution 2.4-17 b)

impulse Response $h(t) = [2e^{-3t} - e^{-2t}]u(t)$

input $x(t) = e^{-t}u(t)$

Zero state response $y(t) = x(t) * h(t)$

$y_{zsr}(t) = e^{-t}u(t) * [2e^{-3t} - e^{-2t}]u(t)$

↓

$y_{zsr}(t) = 2 \underbrace{[e^{-3t}u(t) * e^{-t}u(t)]}_{(1)} - \underbrace{[e^{-2t}u(t) * e^{-t}u(t)]}_{(2)}$

final answer: $y(t) = (e^{-t} - e^{-3t})u(t) - (e^{-t} - e^{-2t})u(t) = [e^{-t} - e^{-3t} - e^{-t} + e^{-2t}]u(t) = (e^{-2t} - e^{-3t})u(t)$

we use row 4 of the convolution table, because it matches to our signals. Row 4 says:

$$e^{l_1 t} u(t) * e^{l_2 t} u(t) = \frac{e^{l_1 t} - e^{l_2 t}}{l_1 - l_2} u(t)$$

① $2[e^{-3t}u(t) * e^{-t}u(t)]$, $l_1 = -3$, $l_2 = -1$

$$2 \frac{(e^{-3t} - e^{-t})u(t)}{-3 - (-1)} = 2 \frac{(e^{-3t} - e^{-t})u(t)}{-2} = -(e^{-3t} - e^{-t})u(t) = (e^{-t} - e^{-3t})u(t)$$

② $[e^{-2t}u(t) * e^{-t}u(t)]$, $l_1 = -2$, $l_2 = -1$

$$\frac{e^{-2t} - e^{-t}}{-2 - (-1)} u(t) = -(e^{-2t} - e^{-t})u(t) = (e^{-t} - e^{-2t})u(t)$$

Table 2.1 – Page 176

TABLE 2.1 Select Convolution Integrals

No.	$x_1(t)$	$x_2(t)$	$x_1(t) * x_2(t) = x_2(t) * x_1(t)$
1	$x(t)$	$\delta(t - T)$	$x(t - T)$
2	$e^{\lambda t}u(t)$	$u(t)$	$\frac{1 - e^{\lambda t}}{-\lambda} u(t)$
3	$u(t)$	$u(t)$	$tu(t)$
4	$e^{\lambda_1 t}u(t)$	$e^{\lambda_2 t}u(t)$	$\frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_1 - \lambda_2} u(t) \quad \lambda_1 \neq \lambda_2$
5	$e^{\lambda t}u(t)$	$e^{\lambda t}u(t)$	$te^{\lambda t}u(t)$
6	$te^{\lambda t}u(t)$	$e^{\lambda t}u(t)$	$\frac{1}{2}t^2e^{\lambda t}u(t)$
7	$t^N u(t)$	$e^{\lambda t}u(t)$	$\frac{N! e^{\lambda t}}{\lambda^{N+1}} u(t) - \sum_{k=0}^N \frac{N! t^{N-k}}{\lambda^{k+1} (N - k)!} u(t)$
8	$t^M u(t)$	$t^N u(t)$	$\frac{M! N!}{(M + N + 1)!} t^{M+N+1} u(t)$
9	$te^{\lambda_1 t}u(t)$	$e^{\lambda_2 t}u(t)$	$\frac{e^{\lambda_2 t} - e^{\lambda_1 t} + (\lambda_1 - \lambda_2)te^{\lambda_1 t}}{(\lambda_1 - \lambda_2)^2} u(t)$

Table 2.1 – Page 176

10	$t^M e^{\lambda_1 t} u(t)$	$t^N e^{\lambda_2 t} u(t)$	$\frac{M!N!}{(N+M+1)!} t^{M+N+1} e^{\lambda t} u(t)$
11	$t^M e^{\lambda_1 t} u(t)$	$t^N e^{\lambda_2 t} u(t)$	$\sum_{k=0}^M \frac{(-1)^k M!(N+k)! t^{M-k} e^{\lambda_1 t}}{k!(M-k)!(\lambda_1 - \lambda_2)^{N+k+1}} u(t)$ $+ \sum_{k=0}^N \frac{(-1)^k N!(M+k)! t^{N-k} e^{\lambda_2 t}}{k!(N-k)!(\lambda_2 - \lambda_1)^{M+k+1}} u(t)$
	$\lambda_1 \neq \lambda_2$		
12	$e^{-\alpha t} \cos(\beta t + \theta) u(t)$	$e^{\lambda t} u(t)$	$\frac{\cos(\theta - \phi) e^{\lambda t} - e^{-\alpha t} \cos(\beta t + \theta - \phi)}{\sqrt{(\alpha + \lambda)^2 + \beta^2}} u(t)$ $\phi = \tan^{-1}[-\beta/(\alpha + \lambda)]$
13	$e^{\lambda_1 t} u(t)$	$e^{\lambda_2 t} u(-t)$	$\frac{e^{\lambda_1 t} u(t) + e^{\lambda_2 t} u(-t)}{\lambda_2 - \lambda_1} \quad \text{Re } \lambda_2 > \text{Re } \lambda_1$
14	$e^{\lambda_1 t} u(-t)$	$e^{\lambda_2 t} u(-t)$	$\frac{e^{\lambda_1 t} - e^{\lambda_2 t}}{\lambda_2 - \lambda_1} u(-t)$

Question 2.4-21

2.4-21 A first-order allpass filter impulse response is given by

$$h(t) = -\delta(t) + 2e^{-t}u(t)$$

- (a) Find the zero-state response of this filter for the input $e^t u(-t)$.
- (b) Sketch the input and the corresponding zero-state response.

Solution 2.4-21 a)

Impulse Response $h(t) = -\delta(t) + 2e^{-t}u(t)$

input $x(t) = e^t u(-t)$

Zero state response $y(t) = x(t) * [-\delta(t) + 2e^{-t}u(t)]$

$$\downarrow$$

$$y_{zsr}(t) = \underbrace{-(x(t) * \delta(t))}_{(1)} + \underbrace{2[x(t) * e^{-t}u(t)]}_{(2)}$$

Final answer $y_{zsr}(t) = -\cancel{e^t u(-t)} + \cancel{e^{-t} u(t)} + \cancel{e^t u(-t)} =$

$$y_{zsr}(t) = e^{-t} u(t)$$

① For the first Part we should use Row 1 of the table:

Row 1 says: $x(t) * \delta(t-T) = x(t-T)$

here in our first Part (T) is zero so

$$-x(t) * \delta(t-0) = -x(t-0) \rightarrow -x(t) * \delta(t) = -x(t) = \boxed{-e^t u(-t)}$$

② For the second Part we use Row 13 of the table.

Row 13 says: $e^{l_1 t} u(t) * e^{l_2 t} u(t) = \frac{e^{l_1 t} u(t) + e^{l_2 t} u(t)}{l_2 - l_1}$

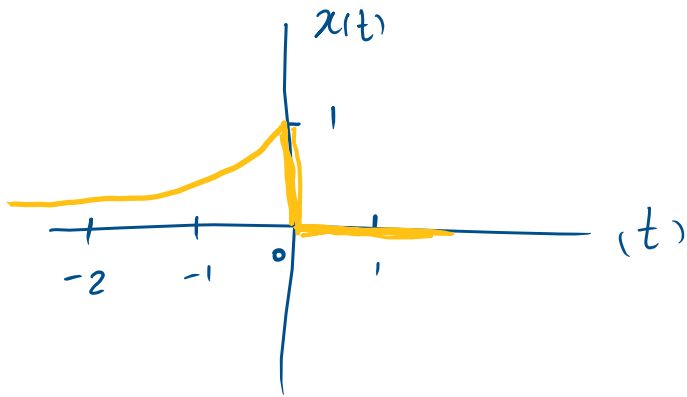
$$2[e^t u(-t) * e^{-t} u(t)] = 2 \left[\frac{e^{-t} u(t) + e^t u(-t)}{\cancel{l_2 - l_1}} \right] = \boxed{e^{-t} u(t) + e^t u(-t)}$$

$\downarrow \quad \quad \downarrow$
 $l_2 = 1 \quad \quad l_1 = -1$

Solution 2.4-21

b)

input $x(t) = e^t u(-t) \rightarrow x(t) = \begin{cases} e^t, & t \leq 0 \\ 0, & t > 0 \end{cases}$



Zero step response $y(t) = e^{-t} u(t) \rightarrow y(t) = \begin{cases} 0, & t < 0 \\ e^{-t}, & t \geq 0 \end{cases}$

* Unit step function $u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$

now test 3 points to sketch it

$t = 0 \rightarrow e^0 = 1$

$t = -1 \rightarrow e^{-1} = 0.37$

$t = -2 \rightarrow e^{-2} = 0.14$

$t = 0 \rightarrow e^0 = 1$

$t = 1 \rightarrow e^{-1} = 0.37$

$t = 2 \rightarrow e^{-2} = 0.14$

